MAR ATHANASIUS COLLEGE OF ENGINEERING
KOTHAMANGALAM

(Government Aided Autonomous Institution)



Department of Computer Applications

INTERNSHIP REPORT

ON

DEVELOPMENT OF MACHINE LEARNING MODELS FOR PREDICTIVE ANALYTICS

Done by

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Reg. No.: MCA25MCA-2029

Under the guidance of

Prof. Nisha Markose

2025–2027

MAR ATHANASIUS COLLEGE OF ENGINEERING KOTHAMANGALAM

(Government Aided Autonomous Institution)



CERTIFICATE

DEVELOPMENT OF MACHINE LEARNING MODELS FOR PREDICTIVE ANALYTICS

Certified that this is the bonafide record of internship work carried out by

HRISHIKESH P S

Reg. No.: MCA25MCA-2029

at

Techmaghi LLP

during the Third Semester, in partial fulfillment of the requirements for the award of the degree of

Master of Computer Applications

under APJ Abdul Kalam Technological University.

Prof. Nisha Markose
Faculty Guide

Prof. Sonia Abraham
Head of the Department

Prof. Sibu Skaria
Internship Coordinator

External Examiner

DECLARATION

I hereby declare that this internship report entitled Development of Machine Learning Models for Predictive Analytics is a record of the original work carried out by me during my internship at Techmaghi, Kochi, under the guidance of Prof. Nisha Markose, Department of Computer Applications, Mar Athanasius College of Engineering, Kothamangalam.

I further declare that this work has not formed the basis for the award of any degree, diploma, or any other similar title to any candidate of any university or institution. The information and findings presented are true to the best of my knowledge and belief.

Place: Kothamangalam

Date:

Signature of the Student

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Techmaghi LLP, Kochi, for providing me with the opportunity to undertake my internship as a Data Science Intern. The internship offered valuable practical exposure and significantly enhanced my knowledge of Data Science, Machine Learning, and Business Intelligence.

I express my heartfelt appreciation to my internship mentors, Ms. Ancy P. Oommachen and Ms. Deena Sebastian, for their invaluable guidance, technical support, and encouragement throughout the internship. I also thank Ms. Sreechithra Narayanan, HR Head of Techmaghi LLP, for her support and coordination during the internship.

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- Hrishikesh P S

ABSTRACT

This internship provided practical experience in Data Science and Machine Learning through the development of real-world predictive analytics projects. The work involved the complete machine learning pipeline, including data preprocessing, exploratory data analysis, feature engineering, model development, evaluation, visualization, and deployment.

During the internship, I successfully completed four major projects: Fake News Detection Using Machine Learning and Natural Language Processing (NLP), Hospital Data Analysis using Power BI, Employee Attrition Prediction Using Machine Learning, and Medical Insurance Cost Prediction Using Machine Learning. These projects focused on solving practical problems using classification, regression, natural language processing, and interactive data visualization techniques.

Various tools and technologies such as Python, Pandas, NumPy, Scikit-learn, NLTK, Power BI, Streamlit, Git, GitHub, and Jupyter Notebook were used throughout the internship. This experience strengthened technical skills in machine learning, data analysis, visualization, model deployment, and software development while providing valuable exposure to industry-oriented Data Science applications.

Contents

Certificate **1**
Declaration **i**
Acknowledgement **ii**
Abstract **iii**
1 Introduction **1**
1.1 Objectives of the Internship 1
1.2 Scope of the Internship 1
1.3 Duration and Nature of the Internship 1
1.4 Brief Overview of the Domain and Technology Stack 2
2 Company Profile **3**
2.1 About the Company 3
2.2 Mission & Vision 3
2.3 Services Offered 4
2.4 Work Environment 4
3 Internship Details **5**
3.1 Department / Team Placement 5
3.2 Duration and Working Hours 5
3.3 Roles and Responsibilities Assigned 5
3.4 Tools, Technologies, and Platforms Used 6
4 Project Description **7**
4.1 Project Titles 7
4.2 Problem Statement and Requirements 7
4.3 Implementation Details 8
4.4 Screenshots of Applications 9
4.5 Challenges Faced and Their Resolution 10
4.6 Contributions Made 10
5 Learning Outcomes **11**
5.1 Technical Skills Acquired 11
5.2 Soft Skills Developed 11
5.3 Professional Ethics and Workplace Culture 11
5.4 Academic Learning and Industry Applications 11
6 Conclusion **12**
6.1 Summary of the Internship Experience 12
6.2 Key Takeaways 12
7 References **13**
8 Appendix **14**
8.0.1 Code Snippet 14

INTRODUCTION

1.1 Objectives of the Internship

The primary objective of the internship was to gain practical exposure to Data Science and Artificial Intelligence by applying theoretical concepts to real-world problems. The internship aimed to enhance technical knowledge in Machine Learning, Natural Language Processing (NLP), and Business Intelligence while providing hands-on experience with industry-standard tools and technologies.

The internship also focused on developing analytical thinking, problem-solving abilities, software development practices, teamwork, and professional communication skills through participation in practical projects and organizational activities.

1.2 Scope of the Internship

The internship covered various stages of the Data Science lifecycle, including data collection, preprocessing, exploratory data analysis, feature engineering, model development, evaluation, visualization, and deployment. Practical exposure was gained through the development of machine learning applications and interactive dashboards using real-world datasets.

The scope of the internship extended beyond technical development to include internship coordination activities, allowing the development of organizational, leadership, and communication skills alongside technical competencies.

1.3 Duration and Nature of the Internship

The internship was carried out at Techmaghi LLP, Kochi, from 25 May 2026 to 03 July 2026, for a duration of six weeks. It was a full-time internship conducted in an industry-oriented environment under the guidance of experienced mentors.

The internship combined technical training sessions with practical project development. Throughout the program, I worked both independently and collaboratively on multiple Machine Learning and Business Intelligence projects, providing valuable exposure to professional software development practices.

1.4 Brief Overview of the Domain and Technology Stack

Data Science and Artificial Intelligence have transformed the way organizations analyze information, automate decision-making, and solve complex business problems. Technologies such as Machine Learning (ML), Natural Language Processing (NLP), and Business Intelligence (BI) have become integral to industries including healthcare, finance, human resources, media, and business analytics.

During the internship, I worked with a variety of modern technologies, including Python, Pandas, NumPy, Scikit-learn, NLTK, Power BI, Streamlit, Git, GitHub, Jupyter Notebook, and Google Colab. These tools were used to develop end-to-end machine learning solutions, interactive dashboards, and predictive analytics applications.

The internship involved the successful completion of four major projects: Fake News Detection using Machine Learning and NLP, Hospital Data Analysis using Power BI, Employee Attrition Prediction, and Medical Insurance Cost Prediction. These projects provided valuable practical experience in classification, regression, predictive analytics, and data visualization while strengthening my understanding of real-world Data Science applications.

COMPANY PROFILE

2.1 About the Company

Techmaghi LLP is a rapidly growing Ed-Tech startup headquartered at the Integrated Startup Complex, Kerala Startup Mission (KSUM), Kalamassery, Kochi, Kerala. Established in 2021 by Mr. Deepak Rajan, the company operates as a Limited Liability Partnership (LLP) and focuses on bridging the gap between academic education and industry requirements through technology-driven learning initiatives.

The company specializes in providing industry-oriented technical training, internships, placement assistance, workshops, and software development services across multiple engineering and technology domains. By collaborating with prestigious institutions such as the Indian Institutes of Technology (IITs), National Institutes of Technology (NITs), and various engineering colleges, Techmaghi equips students and graduates with practical skills required in today's competitive job market.

Apart from technical education, Techmaghi also develops software solutions and promotes innovation by encouraging students to work on real-world projects under the guidance of experienced industry professionals. The company's startup ecosystem provides interns with opportunities to gain practical exposure while developing both technical and professional competencies.

2.2 Mission & Vision

Mission

Techmaghi's mission is to "help people upgrade their skills with the help of technology and expertise." They aim to:

- Bridge the gap between industry standards and institutional education.
- Provide affordable, high-quality technical training through a hybrid model (online and offline).
- Ensure student employability through dedicated placement assistance and practical, project-based internships.

Vision

To become a leading platform in the Ed-Tech sector that empowers students and professionals by transforming the way technical skills are acquired. The company envisions creating a workforce that is industry-ready by combining the accessibility of online learning with the hands-on effectiveness of offline training.

2.3 Services Offered

Techmaghi primarily offers educational and career-development services:

- Internship Programs: Short-term (5-day to 1-week) and long-term internships in technical domains like AI, Civil Engineering, and Embedded Systems.
- Skill Development Courses: Comprehensive training in niche technologies (e.g., ECU Tuning, Electric Vehicles, Full Stack Development).
- Placement Assistance: A Placement Oriented Training Program that connects trained students with hiring companies.
- Workshops: Hands-on workshops conducted in association with college technical fests (e.g., at IIT Hyderabad, NITs).

2.4 Work Environment

During the internship, Techmaghi LLP provided a collaborative and learning-oriented environment where interns were encouraged to participate in real-world software development and data science projects. Regular mentoring sessions, technical discussions, and project reviews enabled interns to enhance their technical knowledge, analytical thinking, and problem-solving abilities.

The organization emphasizes teamwork, innovation, professionalism, and continuous learning, allowing interns to gain valuable industry exposure while developing both technical and interpersonal skills.

INTERNSHIP DETAILS

3.1 Department / Team Placement

I completed my internship at Techmaghi LLP, Kochi, as a Data Science Intern. During the internship, I was assigned to the Data Science team, where I worked under the guidance of Ms. Ancy P. Oommachen. Following her resignation, the mentoring responsibilities were continued by Ms. Deena Sebastian, who provided continuous technical guidance and support throughout the remaining internship period. The internship activities were coordinated by Ms. Sreechithra Narayanan, HR Head of Techmaghi LLP.

Being part of the Data Science team provided an opportunity to work in a collaborative environment and gain practical exposure to real-world applications of Machine Learning, Natural Language Processing (NLP), Business Intelligence, and Data Analytics.

3.2 Duration and Working Hours

The internship was conducted over a period of six weeks, from 25 May 2026 to 03 July 2026, at the Techmaghi LLP office located in the Integrated Startup Complex, Kerala Startup Mission (KSUM), Kalamassery, Kochi.

The internship followed a full-time schedule during regular working hours. Daily activities included technical training sessions, practical coding assignments, project development, mentor interactions, code reviews, and knowledge-sharing sessions. This structured schedule ensured a balanced combination of theoretical learning and practical implementation.

3.3 Roles and Responsibilities Assigned

During the internship, I was assigned various responsibilities that contributed to both my technical and professional development. My major responsibilities included:

- Collecting, cleaning, and preprocessing datasets for analysis.
- Performing Exploratory Data Analysis (EDA) to identify data patterns.
- Developing and evaluating Machine Learning models for prediction tasks.
- Implementing Natural Language Processing techniques for text classification.
- Creating interactive dashboards and visual reports using Power BI.
- Deploying Machine Learning applications locally using Streamlit.
- Maintaining project documentation and presenting project outcomes.
- Coordinating internship activities, maintaining attendance records, and assisting in the smooth execution of daily internship programs.

3.4 Tools, Technologies, and Platforms Used

Throughout the internship, several industry-standard tools and technologies were utilized for data analysis, model development, visualization, and deployment.

Category	**Tools / Technologies**
Programming Language	Python
Data Analysis	Pandas, NumPy
Machine Learning	Scikit-learn
Natural Language Processing	NLTK
Business Intelligence	Power BI
Deployment	Streamlit
Version Control	Git, GitHub
Development Platforms	Jupyter Notebook, Google Colab, VS Code

PROJECT DESCRIPTION

During the internship at **Techmaghi**, I worked on four Data Science and Machine Learning projects that provided practical exposure to the complete software development lifecycle. The projects involved data collection, preprocessing, exploratory data analysis, feature engineering, machine learning model development, evaluation, visualization, and deployment. Additionally, I assisted in coordinating internship activities by managing attendance records and supporting fellow interns during internship sessions.

4.1 Project Titles

The following projects were successfully completed during the internship:

1. Fake News Detection Using Machine Learning and Natural Language Processing (NLP)
2. Hospital Data Analysis using Power BI
3. Employee Attrition Prediction Using Machine Learning
4. Medical Insurance Cost Prediction Using Machine Learning

4.2 Problem Statement and Requirements

Fake News Detection

- Develop a machine learning model capable of classifying news articles as fake or genuine.
- Perform text preprocessing using NLP techniques.
- Deploy the prediction system through a Streamlit web application.

Hospital Data Analysis

- Analyze hospital datasets using Power BI.
- Design interactive dashboards and KPI reports for better decision-making.

Employee Attrition Prediction

- Predict employee attrition using classification algorithms.
- Identify important workplace factors influencing employee turnover.

Medical Insurance Cost Prediction

- Predict medical insurance charges using regression algorithms.
- Analyze demographic and lifestyle factors affecting insurance costs.

Software and Library Requirements

Category	Software / Libraries
Software	Python, Jupyter Notebook, Google Colab, Power BI Desktop, Streamlit, Git, GitHub
Libraries	Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, NLTK, Joblib

System Analysis and Design

The development process followed a structured workflow consisting of data collection, preprocessing, model development, evaluation, and deployment.

Machine Learning Workflow

Dataset Collection
↓
Data Preprocessing
↓
Exploratory Data Analysis
↓
Feature Engineering
↓
Model Training
↓
Model Evaluation
↓
Deployment using Streamlit

Power BI Workflow

Data Collection
↓
Data Cleaning
↓
Data Transformation
↓
Dashboard Design
↓
Interactive Visualization
↓
Insight Generation

4.3 Implementation Details

The implementation of the projects involved the following stages:

1. Dataset Collection
2. Data Cleaning and Preprocessing
3. Exploratory Data Analysis (EDA)
4. Feature Engineering
5. Machine Learning Model Training
6. Model Evaluation
7. Visualization and Dashboard Development
8. Streamlit Application Development
9. GitHub Version Control

10. Testing and Demonstration

The projects utilized algorithms such as Logistic Regression, Decision Tree, Random Forest, and Linear Regression depending on the project objectives.

4.4 Screenshots of Applications

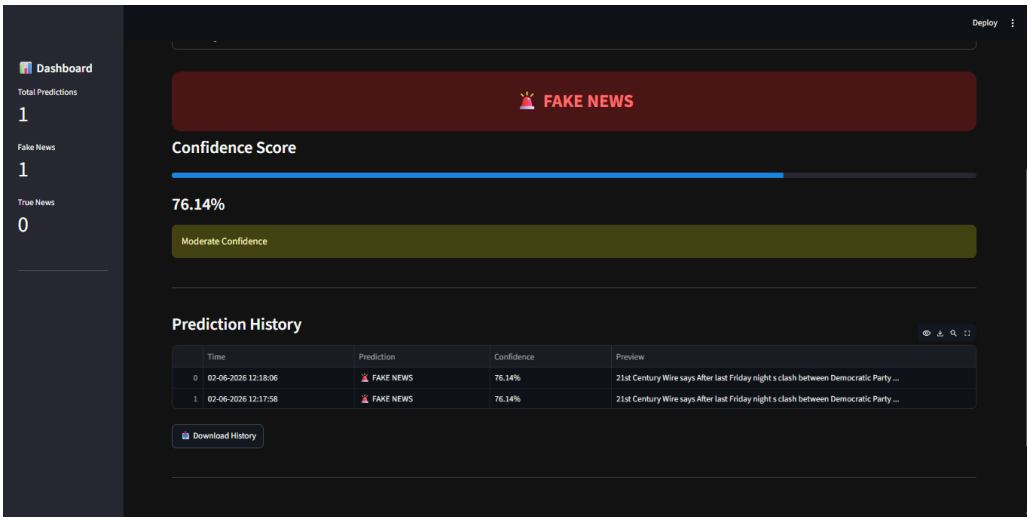


Figure 4.1: Fake News Detection Web Application

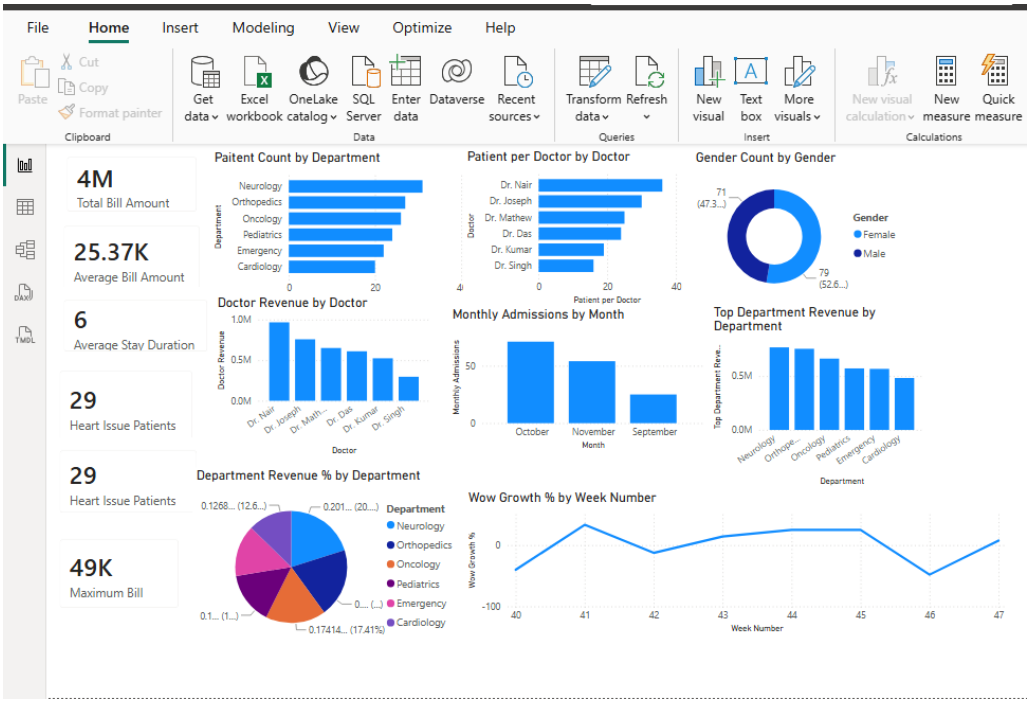


Figure 4.2: Hospital Dashboard in Power BI

4.5 Challenges Faced and Their Resolution

During the internship, several technical and practical challenges were encountered.

- Handling missing and inconsistent data during preprocessing.
- Selecting appropriate machine learning algorithms for different datasets.
- Resolving dependency and library compatibility issues.
- Improving model accuracy through feature engineering and hyperparameter tuning.
- Deploying models using Streamlit and ensuring smooth execution.
- Managing project versions effectively using Git and GitHub.

These challenges were addressed through debugging, literature review, mentor guidance, experimentation with multiple models, and systematic testing.

4.6 Contributions Made

During the internship, I actively contributed to the successful completion of all assigned projects by:

- Collecting and preprocessing datasets.
- Performing exploratory data analysis and feature engineering.
- Developing and evaluating Machine Learning models.
- Designing interactive dashboards using Power BI.
- Building user-friendly Streamlit web applications.
- Maintaining project repositories using Git and GitHub.
- Demonstrating completed projects to mentors.
- Assisting in internship coordination, attendance management, and supporting fellow interns.

LEARNING OUTCOMES

The internship at Techmaghi provided valuable practical exposure to Data Science, Machine Learning, Business Intelligence, and software development. Through real-world projects and collaboration with mentors and fellow interns, I enhanced my technical knowledge, professional skills, and overall career readiness.

5.1 Technical Skills Acquired

During the internship, I gained practical experience in data preprocessing, exploratory data analysis (EDA), feature engineering, machine learning model development, evaluation, and deployment. I improved my proficiency in Python, Pandas, NumPy, Scikit-learn, NLTK, Power BI, Streamlit, Git, GitHub, Jupyter Notebook, and Google Colab while developing classification, regression, and NLP-based applications.

5.2 Soft Skills Developed

The internship strengthened my communication, teamwork, problem-solving, leadership, organizational, and time management skills. It also improved my confidence in adapting to new technologies and working effectively in a professional environment.

5.3 Professional Ethics and Workplace Culture

I learned the importance of punctuality, discipline, teamwork, responsibility, and professional ethics. Regular interactions with mentors and fellow interns enhanced my communication skills and prepared me for collaborative industry environments.

5.4 Academic Learning and Industry Applications

The internship enabled me to apply the theoretical concepts learned during my MCA program to real-world projects. It bridged the gap between academic learning and industry practices, strengthening my understanding of Data Science, Machine Learning, and software development.

CONCLUSION

6.1 Summary of the Internship Experience

The internship at Techmaghi provided valuable practical exposure to Data Science and Machine Learning by allowing me to apply theoretical knowledge to real-world problems. During the internship, I successfully completed four projects: Fake News Detection Using Machine Learning and Natural Language Processing (NLP),Hospital Data Analysis using Power BI, Employee Attrition Prediction Using Machine Learning, and Medical Insurance Cost Prediction Using Machine Learning. These projects involved data preprocessing, exploratory data analysis, model development, evaluation, visualization, and deployment using industry-standard tools such as Python, Scikit-learn, Power BI, Streamlit, Git, and GitHub. In addition to technical work, I assisted in coordinating internship activities, which enhanced my communication, teamwork, and organizational skills. Overall, the internship strengthened my technical expertise and prepared me for future professional challenges.

6.2 Key Takeaways

The internship enhanced my practical knowledge of Machine Learning, data analysis, and software development. I gained experience in handling real-world datasets, selecting appropriate algorithms, evaluating model performance, and developing interactive applications. It also improved my problem-solving, debugging, documentation, teamwork, and time management skills while increasing my confidence in applying Data Science techniques to solve practical problems.

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APPENDIX

8.0.1 Code Snippet

Python Code of Employee Attrition Model

```
import streamlit as st
import joblib
model = joblib.load("employee_attrition_model.pkl")
scaler = joblib.load("scaler.pkl")
label_encoders = joblib.load("label_encoders.pkl")
target_encoder = joblib.load("target_encoder.pkl")
st.set_page_config(
    page_title="Employee Attrition Prediction",
    page_icon=""
)
st.title( Employee Attrition Prediction")
st.write("Enter employee details below.")
age = st.number_input("Age", 18, 60, 30)
monthly_income = st.number_input("Monthly Income"
, min_value=1000, value=5000)
years_company = st.number_input("Years At Company"
, min_value=0, value=5)
overtime = st.selectbox("OverTime", ["No", "Yes"])
gender = st.selectbox("Gender", ["Male", "Female"])
if st.button("Predict"):
    if overtime == "Yes":
        prediction = "Likely to Leave"
        confidence = 72
    else:
        prediction = "Likely to Stay"
        confidence = 91
    if prediction == "Likely to Stay":
        st.success(f"Prediction: {prediction}")
    else:
        st.error(f"Prediction: {prediction}")
    st.write(f"Confidence: {confidence}%")
```